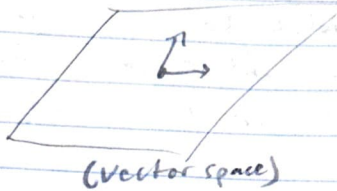


Lecture 10-16

Stratatz 86: Quasi-periodicity

2D (+1 dimension of time)

2D cartesian plane



all spaces have well defined metric



cylinder



torus

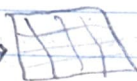
(cylinder w periodic BCs in both directions)

A topological surface of genus 1

torus

crystal

unwrap



→ The torus domain is a natural choice for periodically structured objects, like crystals (e.g., graphene)

→ Periodicity: a system solution with one coordinate that satisfies $q(t) = q(t+2\pi)$

→ Quasi-Periodicity: when a system admits solutions with multiple, concurrent periodicities in multiple coordinates & on multiple timescales

→ (G.T. Finkelstein) Penrose, Dirac, Feynman etc considered a universe that wasn't flat/cartesian

→ Surface of genus 2:

Gaussian curvature has consequences for QM, condensed matter using this surface:

→ "a direction where our course doesn't go, but the science went"



→ Strogatz beat Predrag in a faculty search at Cornell 101

→ Strogatz's book does not emphasize the global nature of studying dynamics: i.e., the boundary conditions, or what happens to the coordinate surface @ ∞ isn't a focus. But this is extremely important in modern dynamics

→ If you have a symmetry: Use It!

→ Eugene Wagner: Group theory book

→ To utilize symmetries in nonlinear systems, you need to employ advanced geometric methods

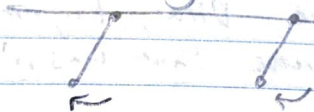
Coupled Oscillators

→ Huygens, clocks

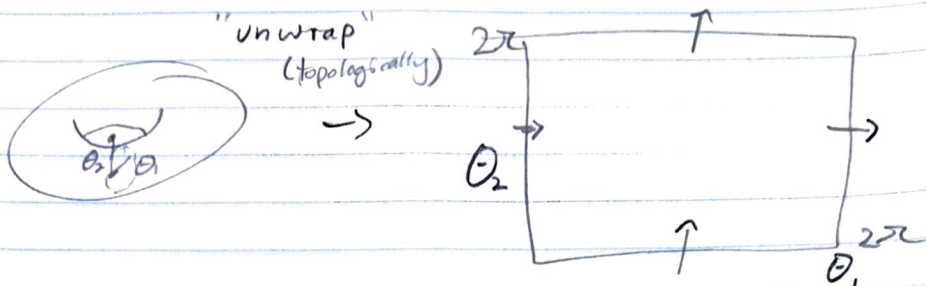
→ clocks (pendulums)

hanging from the same board

↳ coupled by motion



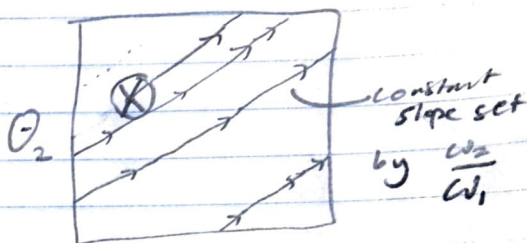
$$\Rightarrow \begin{cases} \dot{\theta}_1 = \omega_1 + k_1 \sin(\theta_2 - \theta_1) \\ \dot{\theta}_2 = \omega_2 + k_2 \sin(\theta_1 - \theta_2) \end{cases}$$



$$\Pi_2 = \{ \theta_i \in [0, 2\pi], i=1,2 \}$$

$\rightarrow$ The coupled system is challenging,
so we turn to the straightforward uncoupled system:

$$\begin{cases} \dot{\theta}_1 = \omega_1 \\ \dot{\theta}_2 = \omega_2 \end{cases}$$



$\rightarrow$ eventually, the system will cover the entire torus

with this 1 line $\rightarrow$ Ergodicity θ_1

$\rightarrow$ If $(\omega_1/\omega_2) \in \mathbb{Q}$ (rational), then

$\Rightarrow$ a constant, invariant manifold is formed by the solution for all time

$\rightarrow$ Ergodicity is still very difficult to establish, for N particles in 3D

$$\begin{cases} \dot{\theta}_1 = \omega_1 + k_1 \sin(\theta_2 - \theta_1) \\ \dot{\theta}_2 = \omega_2 + k_2 \sin(\theta_1 - \theta_2) \end{cases}$$

Let $\phi = \theta_1 - \theta_2 \rightarrow$ symmetry

$$\dot{\phi} = \omega_1 - \omega_2 - (k_1 + k_2) \sin \phi$$

(using $\sin(-\phi) = -\sin \phi$)

We use a consequence of symmetry to reduce the 2 variable problem to the difference of the two

leads us to

$\Rightarrow$ Poincaré Sections